PDF DUO

User guide — PDF, electrical diagrams, GRAFCET and floor plans — free software

Version 1.3.1 · Jacques Branas, en collaboration avec Claude.ai

PDF DUO lets you read, fill in and sign PDFs, draw electrical, pneumatic and network diagrams, GRAFCETs and floor plans, and convert Word / ODF ↔ PDF. It is designed for students and teachers in all vocational courses (for example electrical, IT, dental technician, logistics, warehouse), and is also handy for school life: posters, membership forms and signed documents for the student association. The red numbers on the screenshots refer to the lists below them.

1. Installation

Windows: run the installer `PDF-DUO-Windows-Setup.exe`, or unzip `PDF-DUO-Windows.zip` and open `PDF DUO.exe`.

Linux: double-click `pdf-duo_amd64.deb` (Debian, Ubuntu, Mint), or use the AppImage (make it executable, then open it).

macOS: pick the Apple Silicon (M1, M2…) or Intel version, unzip, drag “PDF DUO” to Applications. First launch: right-click › Open.

Word → PDF conversions: PDF DUO uses **LibreOffice** if installed (most faithful, free), otherwise **Microsoft Word**, otherwise its own converter (simplified layout). ODT files need LibreOffice.

2. The window

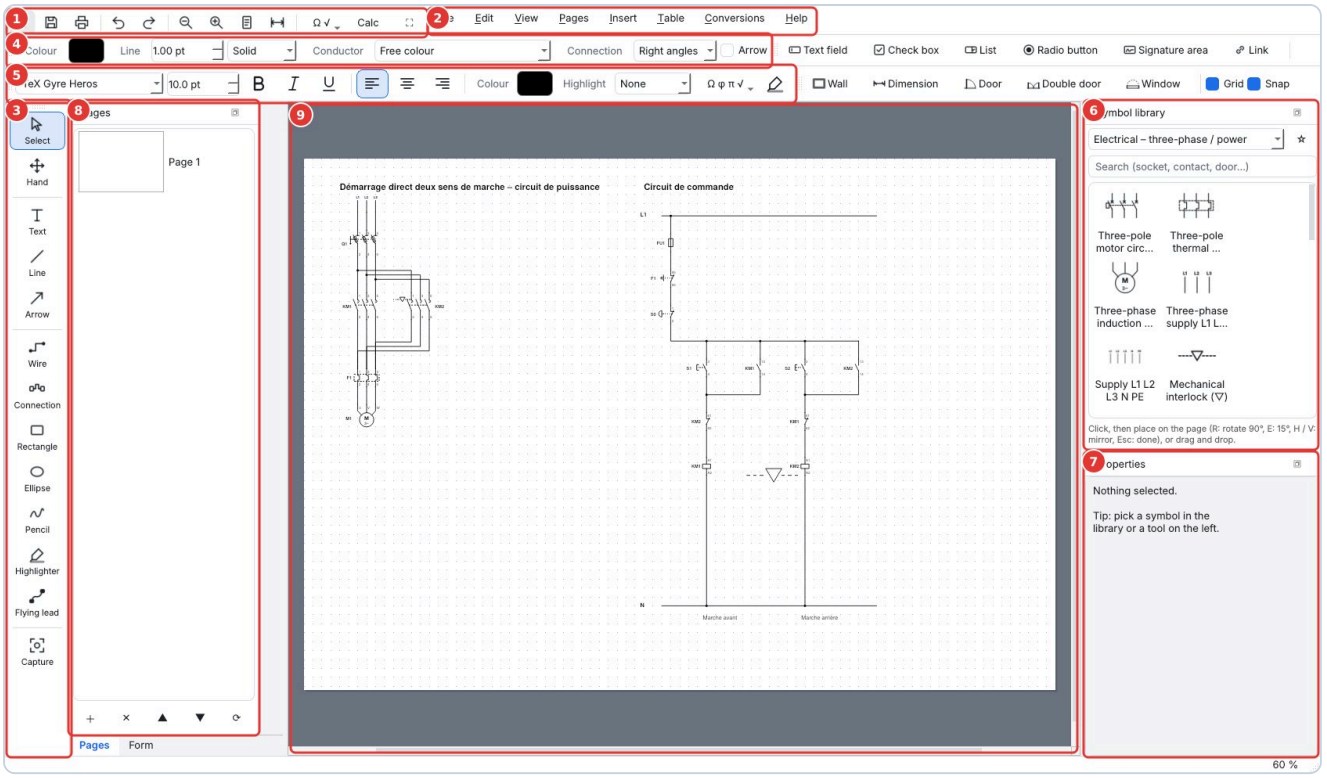
Top, left of the menus: open, save, print, **undo ↶ / redo ↷**, **zoom − / +**, whole page, special characters **Ω √**.

Left: the tools (select, text, line, wire, connection, shapes, pencil, highlighter, flying lead, capture).

Right: the **symbol library** (choose a category or type a word to search) and the **Properties** panel of the selected item.

View › Palettes shows or hides each panel; *View › Theme* and *Language* change the look and language.

Presentation mode (**F11** key or ⛶ button): full screen; only the tools on the left and the library on the right stay visible; menus and zoom reappear when the mouse touches the top of the screen, **Esc** to exit.



The PDF DUO window

1. Quick bar: open, save, print, undo, redo, zoom − / +, Ω √ characters
2. Menus
3. Drawing tools
4. Colour, width, conductor, connection type, arrow
5. Writing palette: font, size, bold, colour, highlighter
6. Symbol library (categories and search)
7. Properties of the selected item
8. Pages (thumbnails) and Form tab
9. The working page

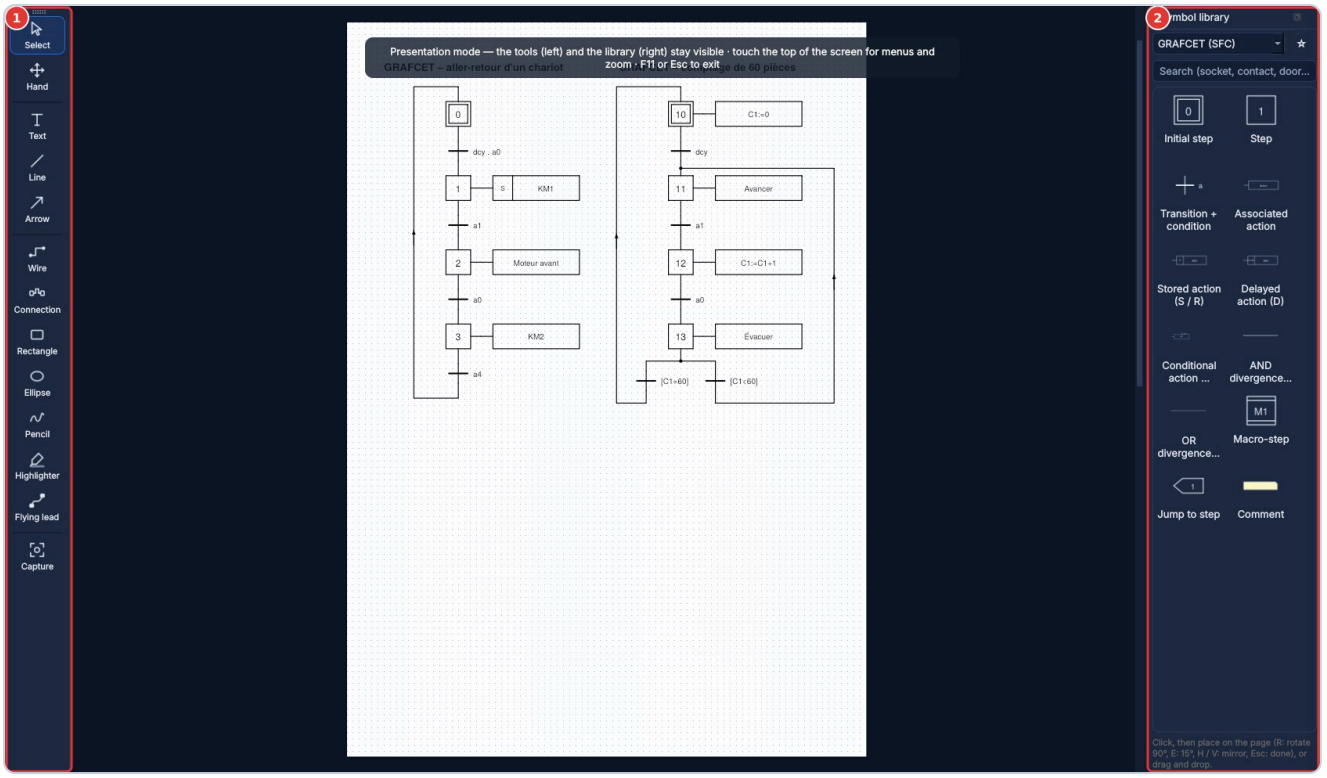
3. Presentation mode (full screen)

To project a diagram in class or work with nothing around, switch to **presentation mode**: the page fills the screen. The toolbar on the left (select, text, line, arrow, wire, shapes, pencil, highlighter) and the library on the right, narrowed to two columns, stay visible: you can still write and place symbols.

To enter it: press **F11**, or the ⛶ button in the top bar, or the menu *View › Presentation mode (full screen)*.

During the presentation: **Ctrl + wheel** to zoom, wheel to scroll. Move the mouse to the very top of the screen: the menus and the quick bar (zoom − / +, whole page, undo) reappear, then hide when the mouse moves down.

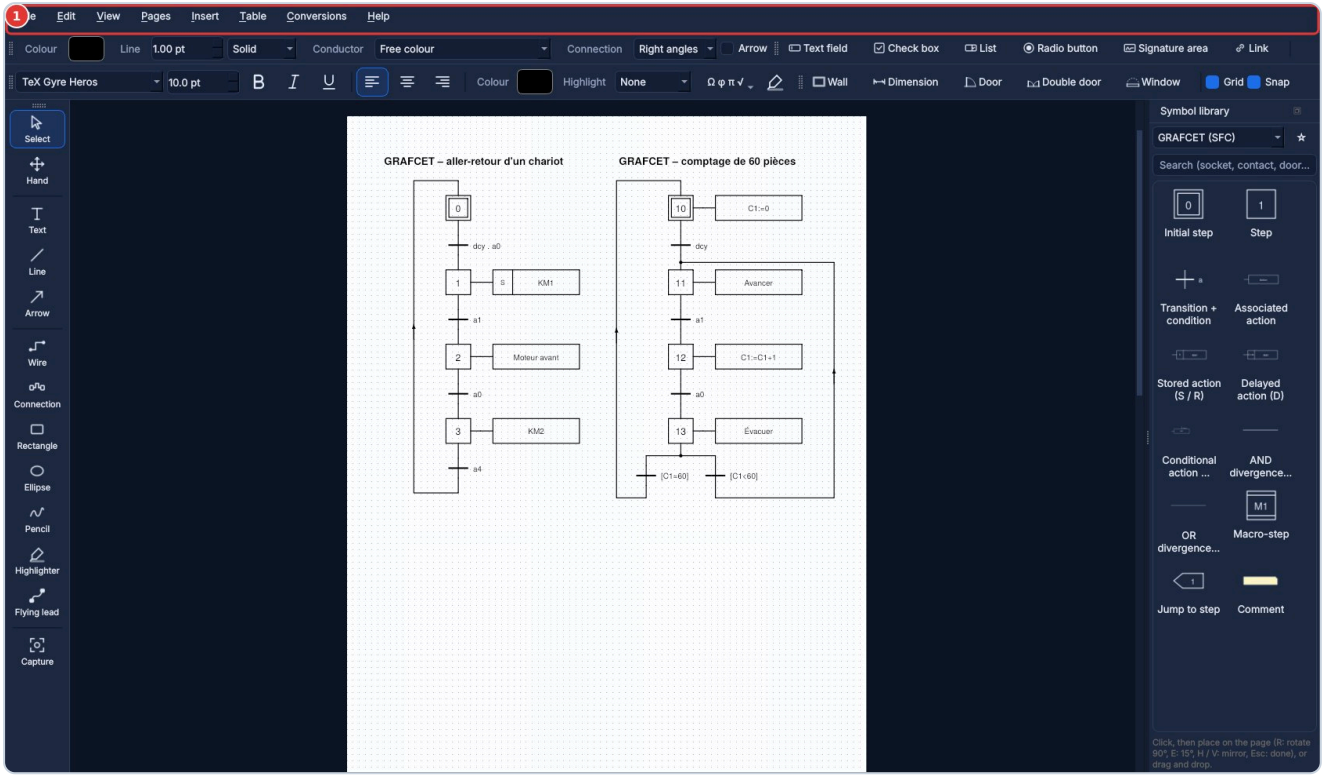
To leave it: press **Esc** (or **F11** again). Palettes and toolbars come back as before. If a drawing tool is in use, the first Esc ends it and the second leaves full screen.



Presentation mode: the page full screen, with the tools and the library

1 Writing and drawing tools

2 Library narrowed to two columns



Mouse at the top of the screen: menus and zoom reappear

1 Menus and quick bar: zoom − / +, whole page, undo, ⛶ to exit

4. Opening and saving

File › Open accepts PDFs, Word / ODT documents (converted automatically) and images.

Drawings are saved as vectors in the PDF. When the file is reopened in PDF DUO, everything stays editable.

File › Export a final PDF produces a flattened PDF to hand out.

5. Drawing a diagram

Click a symbol in the library, then click on the page to place it (or drag and drop). **Esc** to finish.

Favourites: right-click a symbol › *Add to favourites*. It gets a ★ star, moves to the top of its category and appears in the ★ **Favourites** category at the top of the list, which opens at start-up. Right-click › *Remove from favourites* to take it out. Favourites are kept between sessions. The ☆ button next to the list makes a **whole library** a favourite: it moves to the top of the list. **Right-click an item on the page**: bring to front, send to back, rotate, mirror, copy, duplicate, delete, favourites.

R rotates 90°, **E** 15° (Shift = other way), **H** / **V** mirror. Exact angle in Properties.

The **blue handle** at the bottom right enlarges the item proportionally.

The **label** (KM1, S1…) and **terminal numbers** are edited in Properties; “Label on the left / right” buttons.

Wire tool: one click per corner; clicking a terminal (small circle) attaches the wire. **Connection** tool: click two terminals and the route is drawn. Wires follow moved symbols.

Conductor list: standard colours (phase, blue neutral, green/yellow earth, fibre, Ethernet…). Junction dots are placed automatically.

Solar energy: two categories. *Solar energy – symbols* for the standard diagram (PV cell and module, inverter, MPPT / PWM charge controller, 2 / 6 / 12 / 24 / 48 V battery, shunt, DC ammeter and voltmeter, Wh meter, gPV fuse, DC disconnector, surge protector, MC4). *Solar energy – wiring diagram* to draw an off-grid installation with real-looking equipment (panel, charge controller, battery, inverter, monitor, loads), connecting their terminals with red and black wires. Example supplied: `exemple-solaire.pdf`.

Backup sources: wind turbine (with charge controller and dump load), generator set and car or boat alternator, as symbols and realistic pictures. The **2-pole fuse holders** (picture) protect the PV array, the battery output and the load output, as in the example.

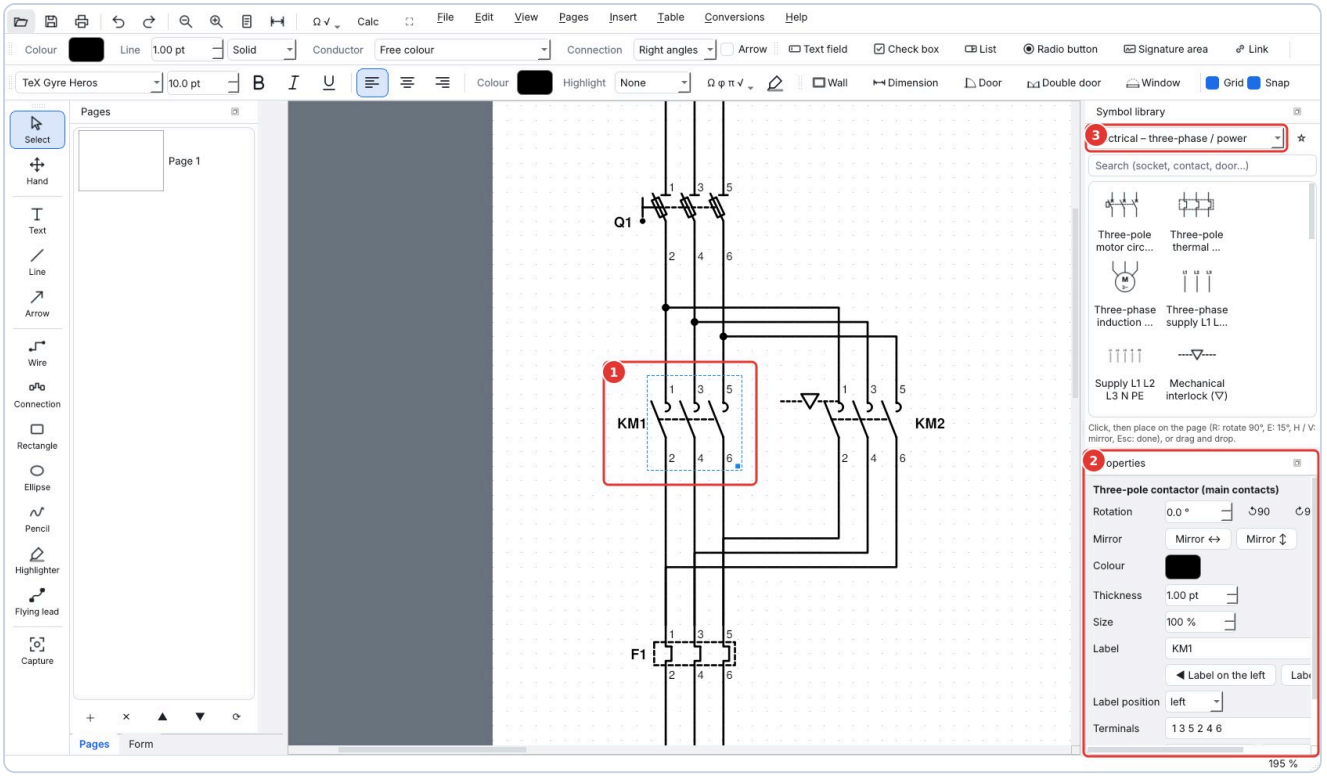
Organisation chart (internship report): *Insert › Company organisation chart* places a ready-linked chart (management, departments, supervisor, intern); double-click a box to type the names. The boxes of the *Organisation chart* category let you extend it.

Libraries added in 1.2.0: **Electrical safety** (PPE, insulating and collective equipment: helmet, insulating gloves, mat, lockout padlock, voltage tester, barriers, earthing device), **Handling** (packaging symbols), **Safety** (signs, GHS hazardous substances), **Logistics**, **Dental technician** and **Tools** for electricians, mechanics and IT (RJ45 crimper, connectors, tester…). Original drawings, free to use. Also: **Working at height** (scaffolding, podium, stepladder, aerial platforms), 1000 V insulated tools, dental chart with FDI numbering and removable dentures.

Libraries added in 1.3.0: **Life-saving actions**, **Basic biology**, **Robotics** (Arduino, Raspberry Pi with GPIO pinout, micro:bit, sensors and modules; adjustable LED colour and resistor bands, colour code chart), **Fire safety** (equipment and signs), **Industrial equipment**, **Housing** (utility duct, heating, lighting, charging points) and **Security careers** (firefighters, security, police).

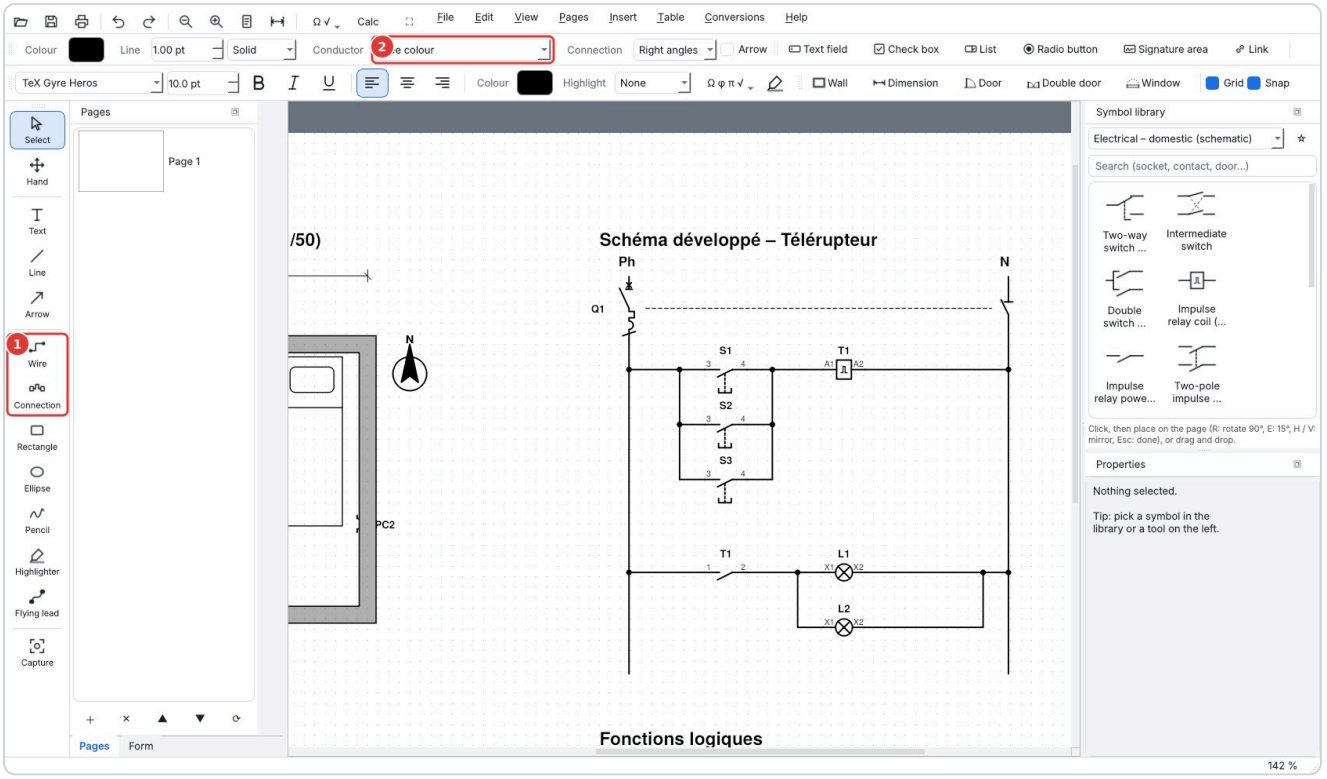
Equipment with states: for the barrier, belt conveyor, hoist, lift, pump, tanks, cylinders, sensors, robot arm… the position or state is set in **Properties** (barrier up, middle or down, water level,

pump running, rod extended…). Each step of a cycle can be shown, for example for a GRAFCET. Examples provided: `exemple-pompage.pdf`, `exemple-parking.pdf`, `exemple-premiers-secours.pdf`.



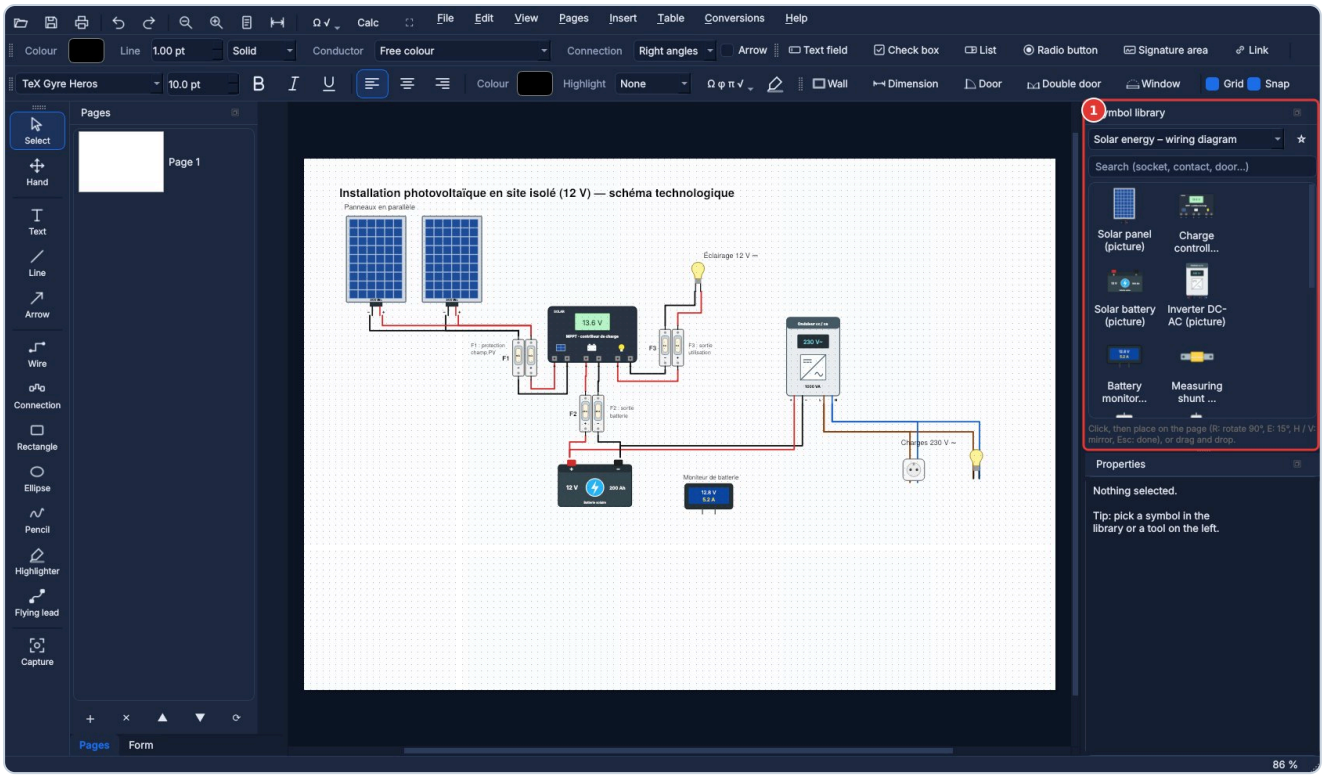
A selected symbol and its properties

1. Selected symbol (dotted frame): blue handle at the bottom right to enlarge it
2. Rotation, mirror, colour, width, size, KM1 label and its position
3. Choice of the symbol category



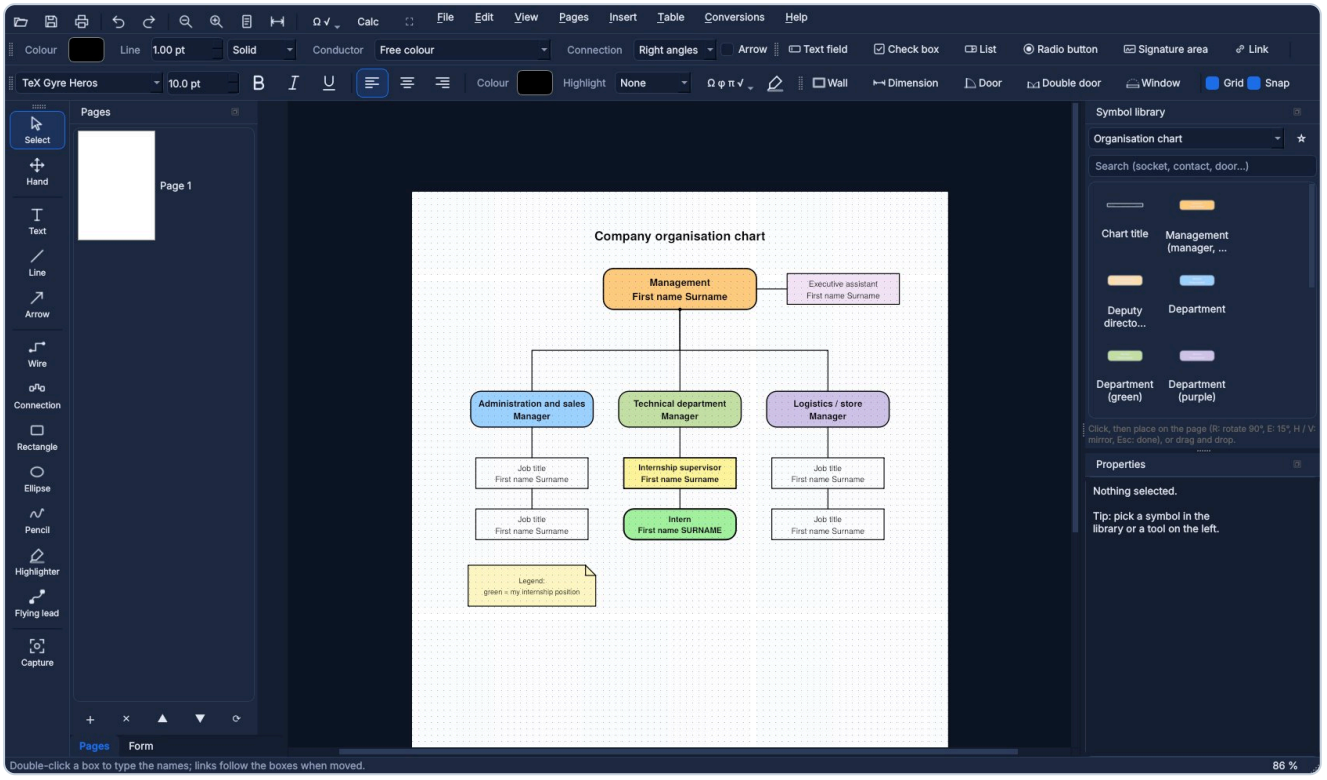
Wires, connections and conductor colours

1. Wire and Connection tools
2. Standard conductor colour

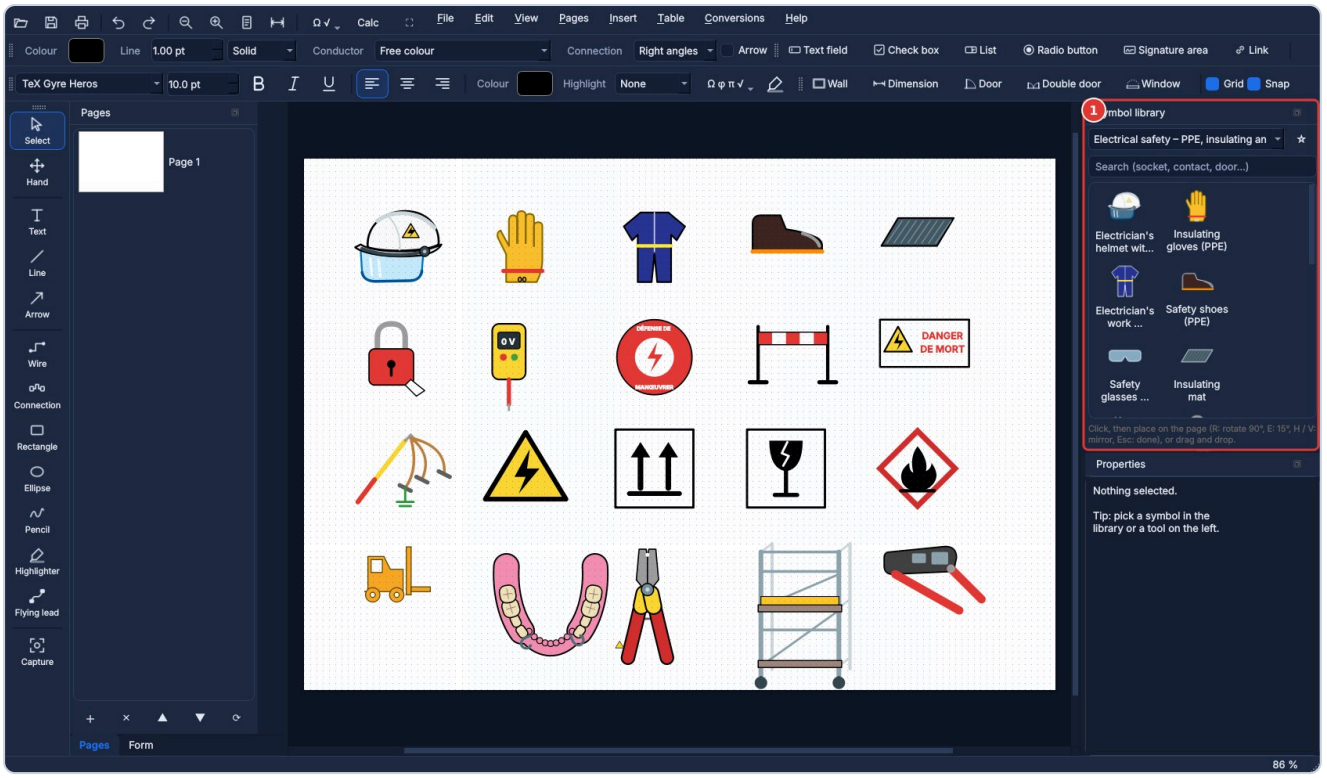


Off-grid solar installation: wiring diagram

1 Solar energy categories of the library

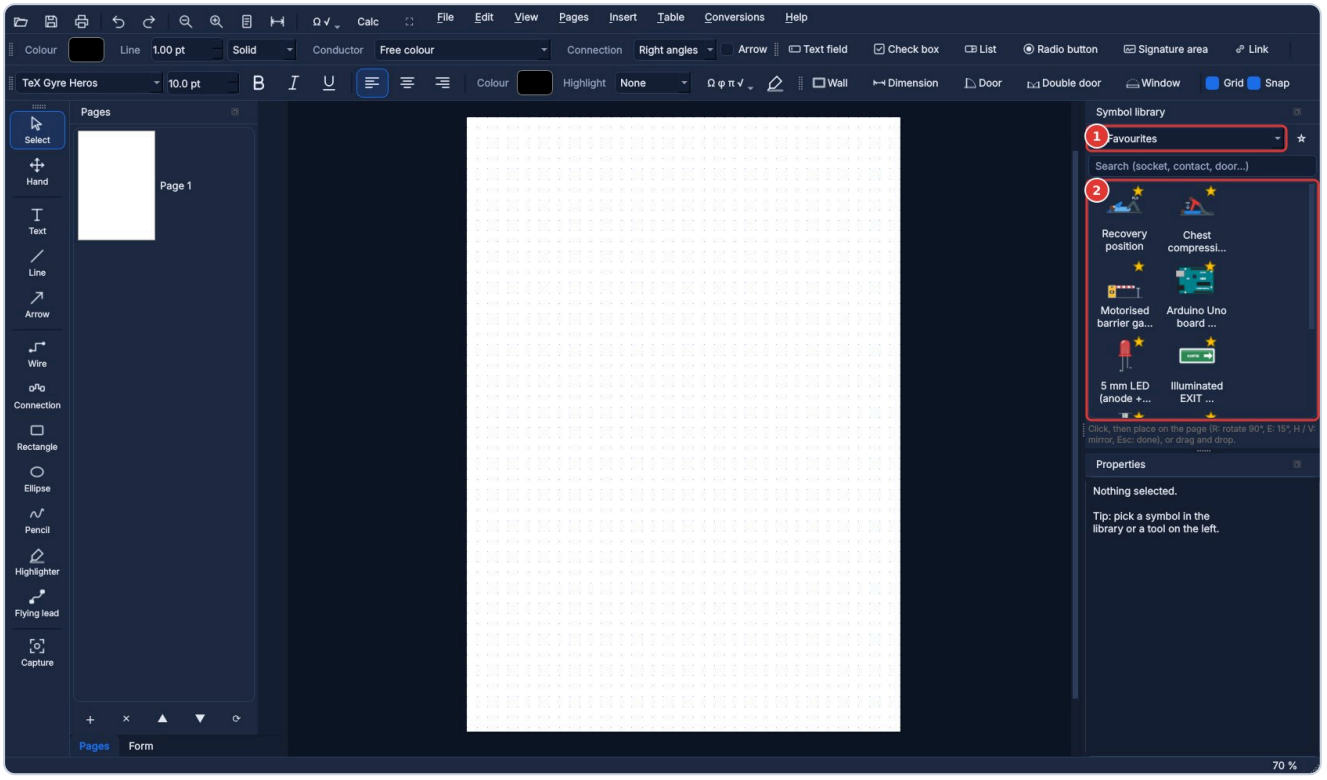


Company organisation chart for the internship report



Electrical safety, handling, logistics, dental, tools

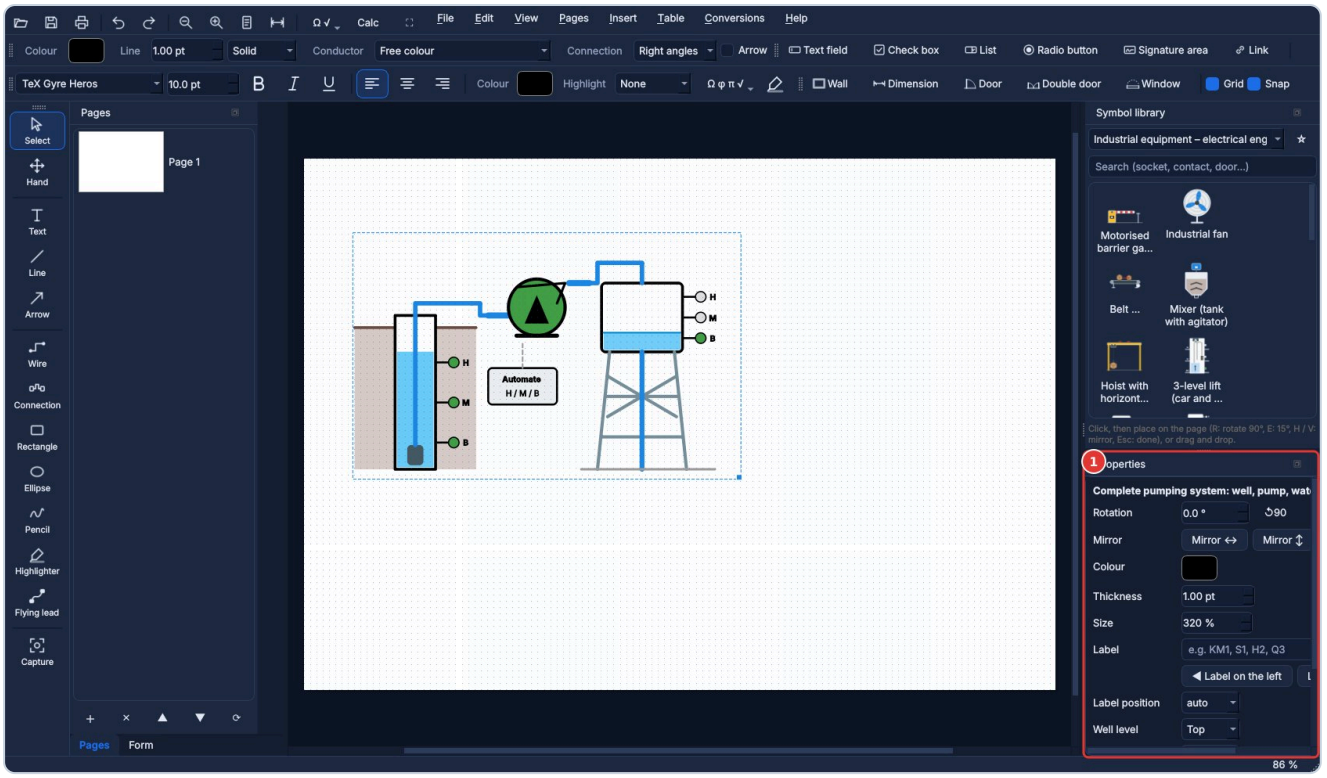
1 Choose the category in the library



Favourites at the top of the library

1 ★ Favourites category

2 Right-click a symbol: add to or remove from favourites



Equipment with states: pumping station, settings in Properties

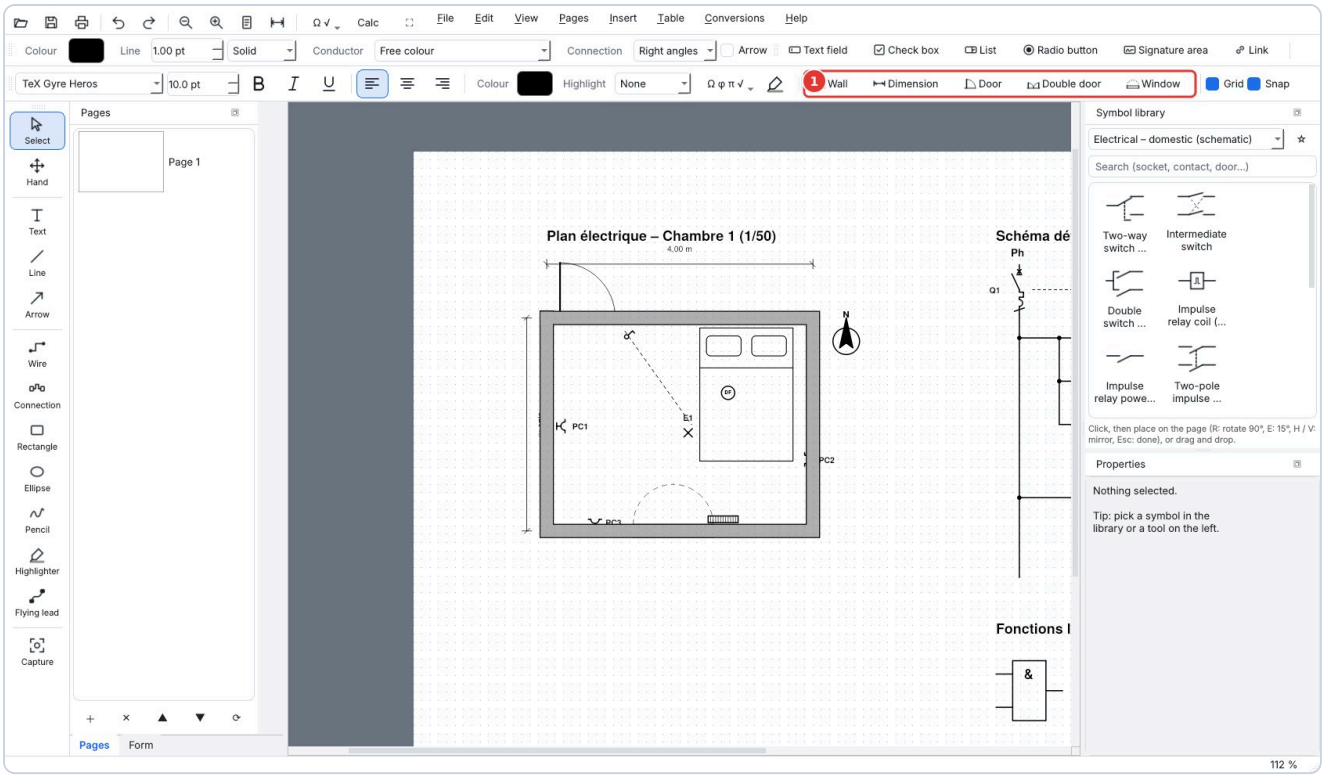
1. Well level, water tower level, pump running or stopped

6. Floor plans

Set the **scale** (1/50, 1/100…) and the real grid step in the *Plan* toolbar.

Wall tool: one click per corner, double-click to finish. **Door**, **Double door**, **Window** tools: move over a wall and the opening aligns and takes the wall thickness.

Dimension tool: drag from point to point, the real length is shown. *Insert › Title block* adds the title block.



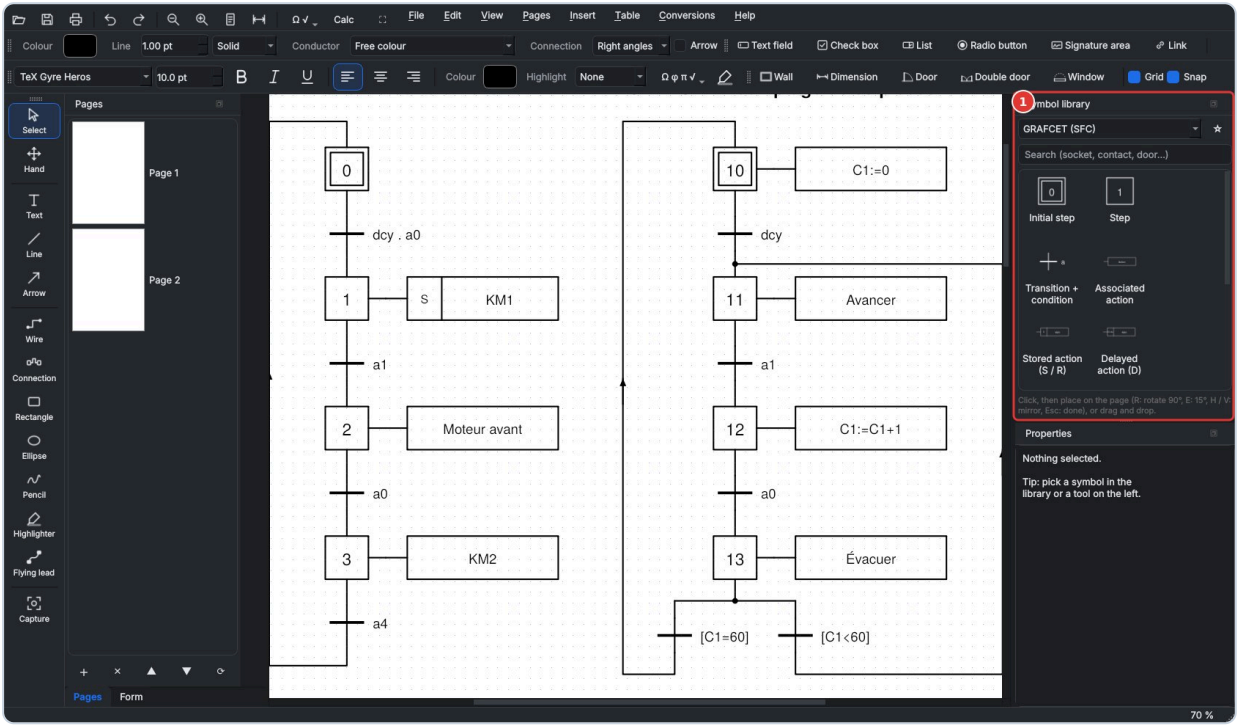
Plan of a bedroom at 1:50: walls, door, window, sockets, dimensions

1. Wall, Dimension, Door, Double door, Window tools

7. GRAFCET, flowcharts, mind maps

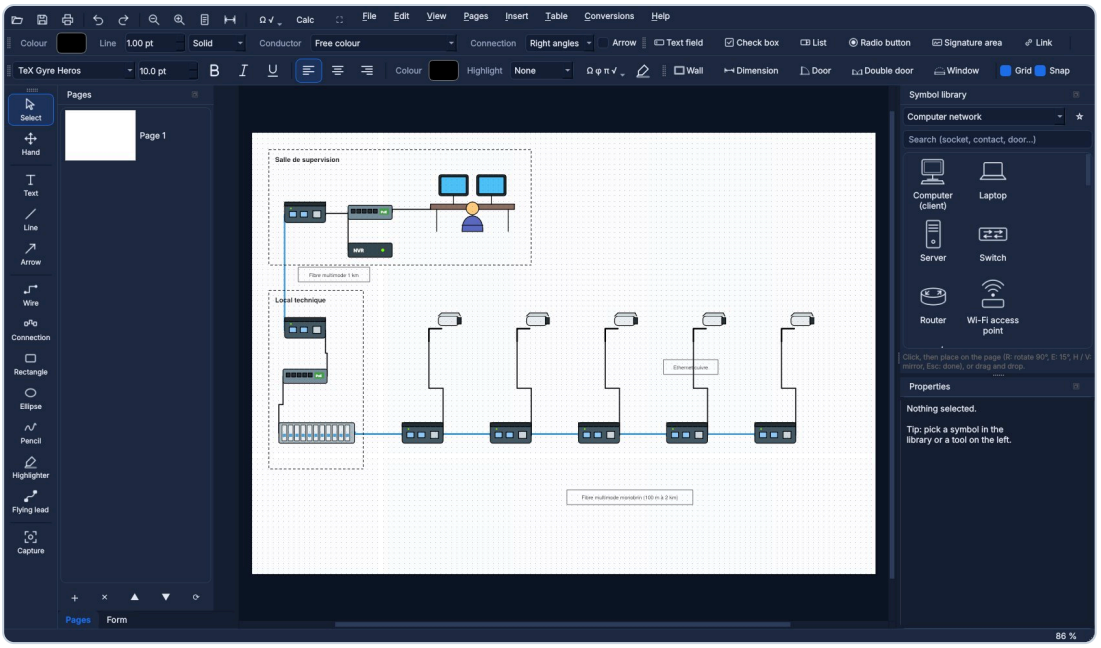
Place an initial step, select it, then press **Ctrl+G**: a transition and the next step are added, connected and numbered.

Double-click a shape to type its text. Choose *Connection: curved* for mind maps, tick *Arrow* for flowcharts.



A GRAFCET: steps, transitions, actions

1. GRAFCET category of the library

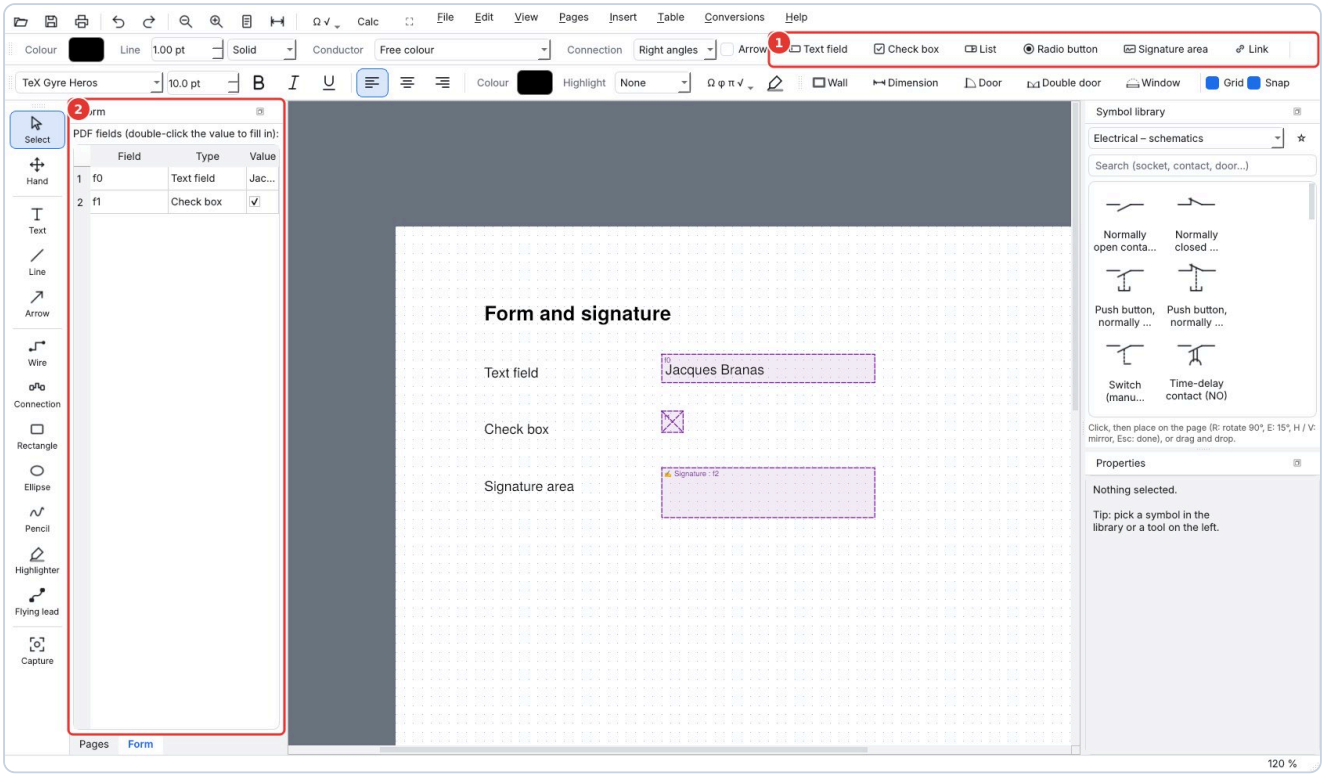


CIEL network diagram (CCTV, fibre) — midnight blue theme

8. Forms and signature

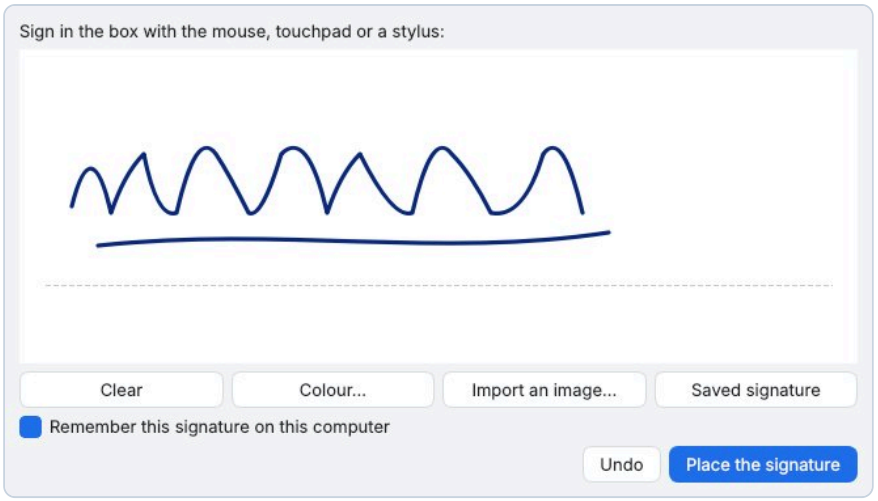
Form and signature toolbar: text field, check box, list, radio button, signature area, link. Double-click to fill in; the *Form* panel lists everything.

Signature button: draw or import your signature, then click where to place it.



Fillable form fields

1. Form and signature toolbar
2. Form panel: all fields and their values



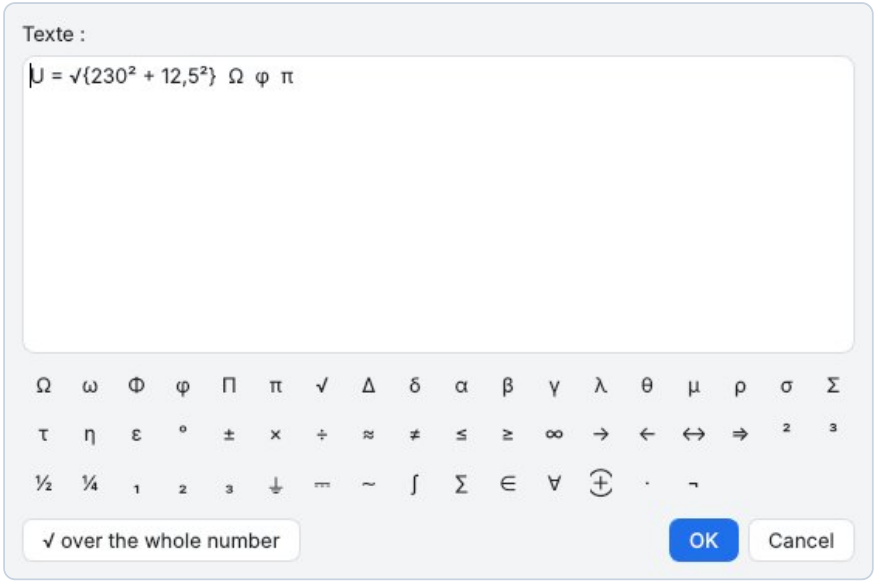
The Signature window

9. Text and special characters

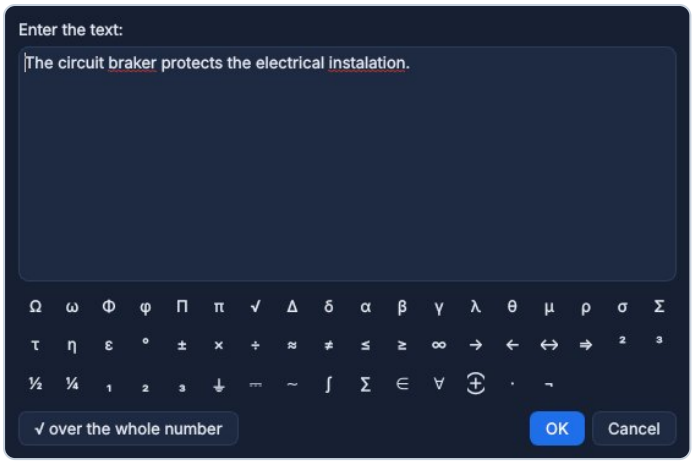
Writing palette: font, size, bold, italic, underline, alignment, colour, highlight.

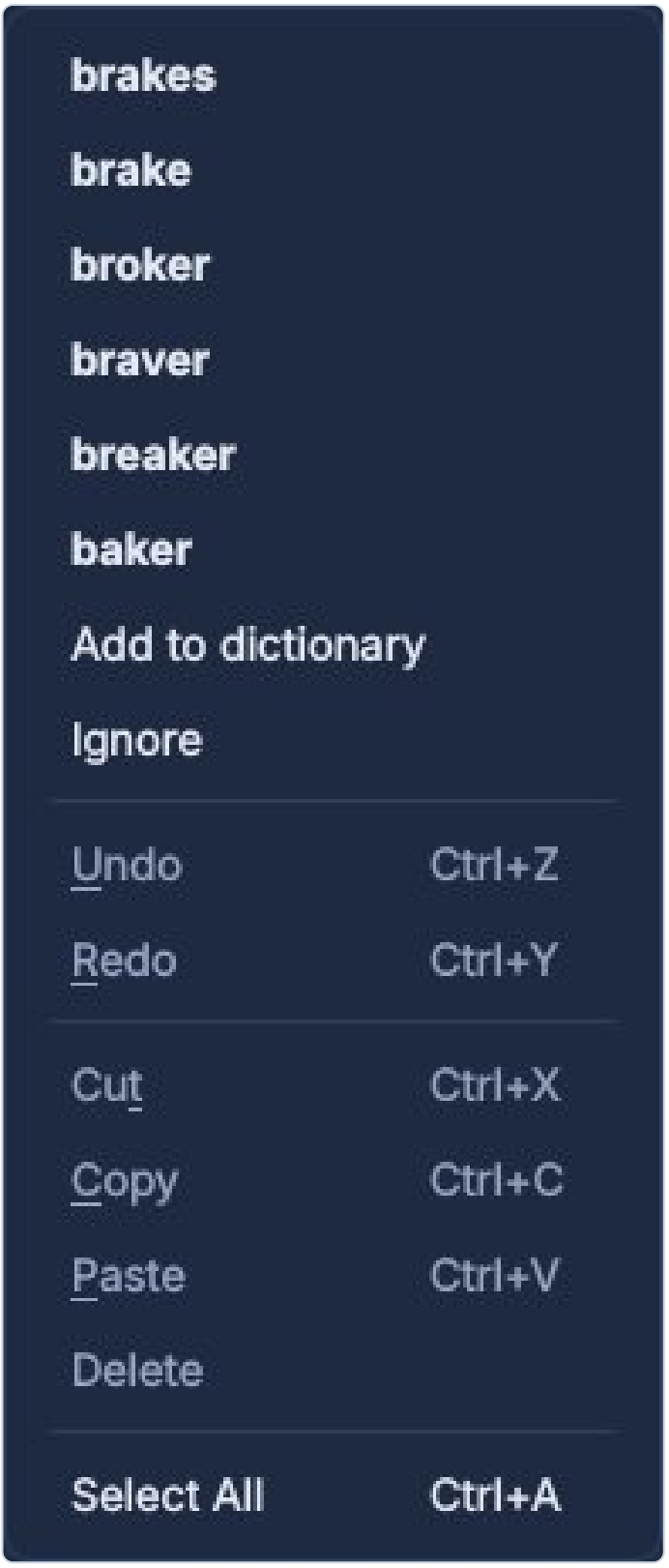
Ω √ button: Ω, φ, π, Δ, μ, °, ±… Type `√{12.5}`: the root bar covers the whole number.

Spell checker: in the text window and in table cells, unknown words are underlined in red. Right-click an underlined word: correction suggestions, *Add to dictionary* (technical terms, proper names) or *Ignore*. Settings in *View › Spell checker*: on or off, language (the program's or any of the six), personal dictionary. Works offline.



Text with special characters and a root over the whole number





Spell checker: underlined words and right-click suggestions

10. Tables and calculator

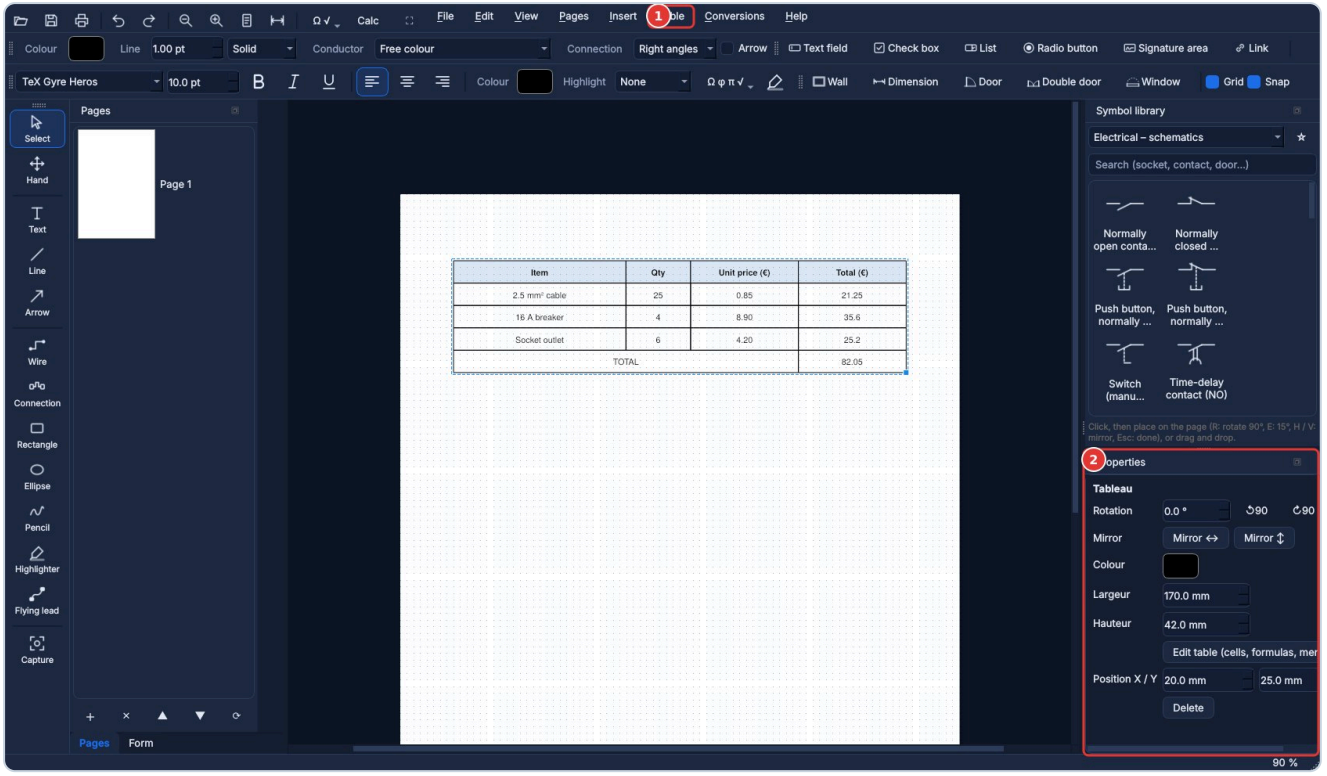
Table › Insert a table… menu (Ctrl+Shift+T): choose the number of rows and columns. The first row can be a header (bold, coloured).

Double-click the table to edit it: add or delete rows and columns, **merge** or **split** cells, set the text size, alignment and header colour. Resize the table with the blue handle.

Spreadsheet-style calculations: a cell starting with **=** is a formula. Examples: `=B2*C2`, `=B3 x C3`, `=(A1+A2)/2`, `=√(144)`, `=A1^2`, `=SUM(D2:D4)`, `=AVERAGE(B2:B6)`, `=MAX(…)`, `=MIN(…)`. The result is shown on the page and updates when the cells change.

Table › Scientific graphing calculator… menu (Ctrl+K): a separate window with sin, cos, tan, ln, log, √, powers, π, in degrees or radians, and a history of calculations. The calculator also opens with the **Calc** button in the top bar.

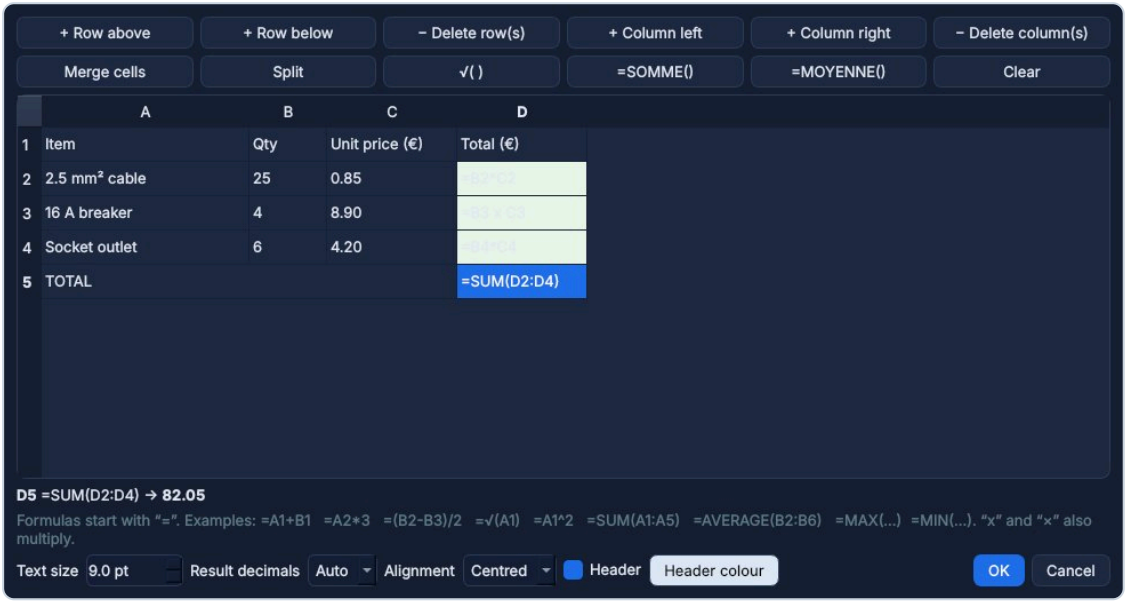
Graph tab: plot up to three curves (e.g. `x^2 - 4`, `2x + 1`, `3sin(x)`), zoom with the wheel, pan by dragging, then **Insert the graph in the page**.



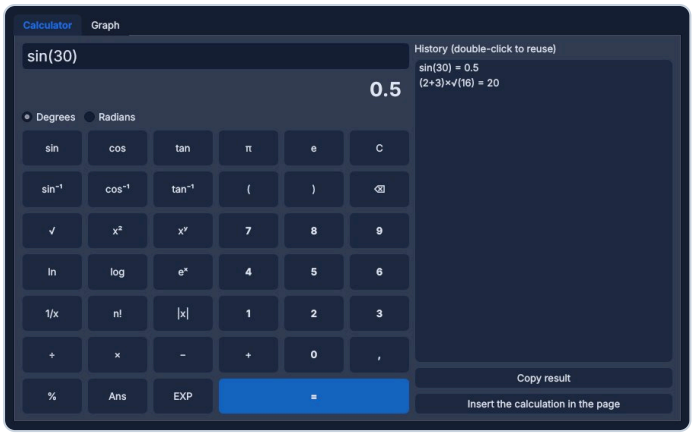
A table with calculations: the totals are formulas

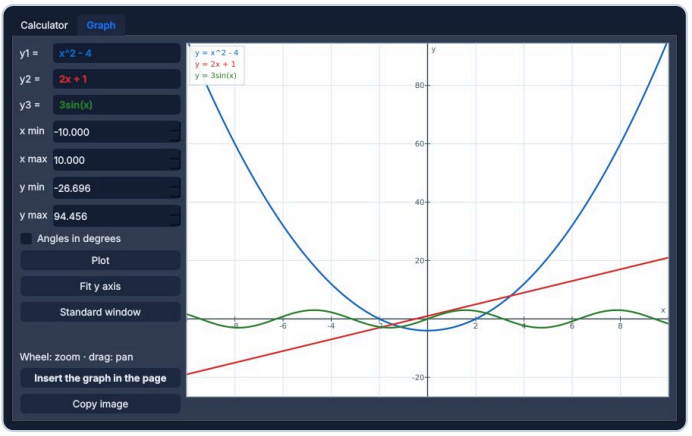
1 Table menu

2 Table size and “Edit table” button



Editing the table (double-click): rows, columns, merge, formulas

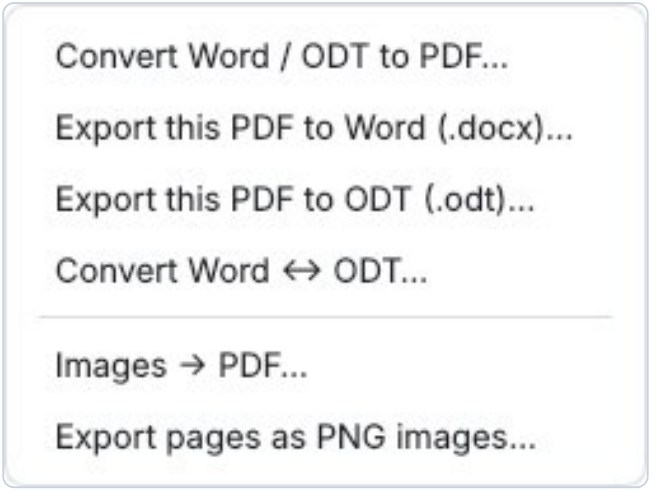




The scientific calculator and its Graph tab

11. Conversions

Conversions menu: Word / ODT → PDF, PDF → Word, PDF → ODT, Word ↔ ODT, images → PDF, pages → PNG.

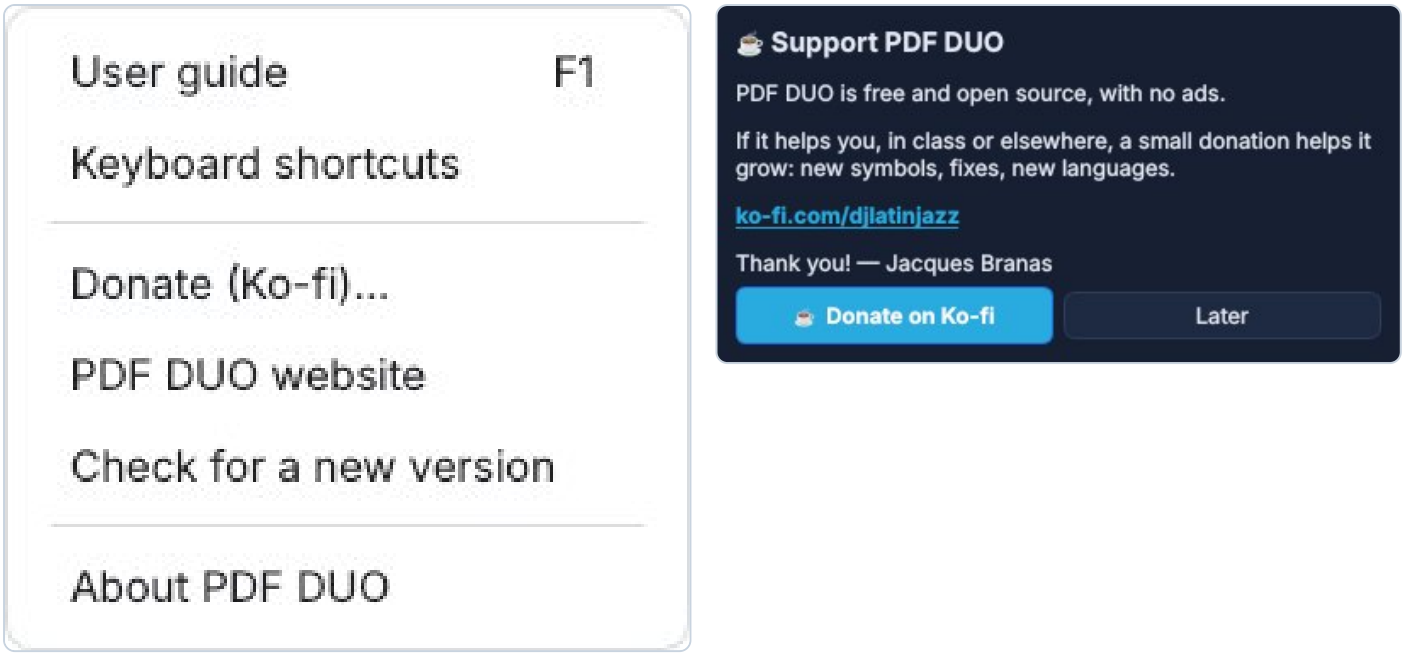


The Conversions menu

12. Support the project

PDF DUO is free software (GNU GPL v3). A donation at **ko-fi.com/djlatinjazz** helps it grow (*Help › Donate*).

Website: deltapassion.github.io/PDF-DUO — Contact: deltapassion@gmx.com



Help menu and donation window